| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2018 | **Task Last Updated:** | FY 05/29/2018 |
| **PI Name:** | Ferl, Robert Ph.D. | | |
| **Project Title:** | Spectral Imaging within The EDEN ISS Project – An Antarctic Analog for Enhancing Exploration Life Support | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | **TechPort:** | | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | (1) Bioregenerative Life Support | | |
| **PI Email:** | robferl@ufl.edu | **Fax:** | FY 352-392-4072 |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 352-392-1928 |
| **Organization Name:** | University of Florida | | |
| **PI Address 1:** | Horticultural Sciences, 1301 Fifield Hall | | |
| **PI Address 2:** | Box 110690 | | |
| **PI Web Page:** | http://www.hos.ufl.edu/ | | |
| **City:** | Gainesville | **State:** | FL |
| **Zip Code:** | 32611-0690 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Space Biology Unsolicited |
| **Start Date:** | 08/15/2017 | **End Date:** | 08/14/2019 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA KSC |
| **Contact Monitor:** | Levine, Howard | **Contact Phone:** | 321-861-3502 |
| **Contact Email:** | howard.g.levine@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Paul, Anna-Lisa Ph.D. ( Co PI: University of Florida ) | | |
| **Grant/Contract No.:** | 80NSSC17K0199 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | This project brings established research from NASA funded science on the International Space Station (ISS) into an international project led by DLR (German Space Agency) toward the functionalization of plant growth in support of human exploration. Imaging technologies developed as part of ongoing ISS science will be deployed within the Antarctic EDEN ISS plant production unit to provide tele-science support and plant science analytics to better understand the responses of plants, and production support to optimize plant performance.<br>Plant production for advanced life support within deep space and extraterrestrial colonies has long been an appreciated area of technology development. Indeed ISS science is now turning toward small scale plant production in space, and plant production units in extreme environments on Earth have proven the value of fresh food. However, there has as yet to be any integrated tests of plant production within space vehicle-like structures in extreme analog environments. To address this gap in knowledge, the EDEN ISS project by the DLR German space agency is designed to place an ISS-inspired module in the context of an established Antarctic station, drawing together for the first time structures and production units designed to space vehicle formats together with the challenges of an extreme and isolated environment. In keeping with the spirit and intent of the EDEN ISS project, the current proposal seeks to fund US participation, bringing plant imaging technologies developed, in part from extensive Shuttle and ISS plant growth experiments, into the science to be conducted within EDEN ISS. The area of fidelity to be addressed by this project is a series of science investigations designed to further planetary research in terrestrial extreme environments that may be analogous to those found on other planets.<br>The EDEN ISS project is constructing essential controlled environment agriculture technologies using both an International Standard Payload Rack cultivation system to relate to on-board ISS and cis-lunar deployments for crew food supplementation, and a Future Exploration Greenhouse that is designed with respect to future planetary bioregenerative life support system deployments. The overall physical facility, referred to as the EDEN ISS Mobile Test Facility, will provide realistic mass flow and will deploy to the highly-isolated Antarctic Neumayer Station III, operated by the Alfred Wegener Institute.<br>Among the technologies to be developed and deployed will be imagers designed by the University of Florida (UF) Principal Investigator (PI) team, through this proposed project, to monitor plant health within the Mobile Test Facility. Our major goal is to equip the plant growth systems with information-rich imaging such that plant health and basic science data can be reviewed by a large and integrated science backroom back on “Earth.” An additional goal is to optimize the conceptual connections between EDEN ISS and current ISS capabilities and operations. These two goals bring a directed focus to the project, connecting the planetary surface analog concepts to near term operations of ISS and potential cis-lunar habitats.<br>To accomplish these goals, spectral imaging of plants will be incorporated into typical data monitoring systems and operation of the project. EDEN ISS, as a large multinational project, has an extensive Data Management Plan (DMP). The work proposed here will depend and draw upon the EDEN ISS DMP for the overall project data, and will specifically draw Mobile Test Facility (MTF) plant growth images and operations images as primary data for analysis and management by the UF team. The UF team will archive and make available all plant health images specific to our tasks via a separate project within the NASA Life Sciences Data Archive (LSDA). The technology is based on small, robust cameras that image wavelengths related to previous and future ISS hardware such that data from EDEN ISS will provide feed forward science for known and anticipated ISS and habitat plant growth hardware. | | |
| **Rationale for HRP Directed Research:** | | | |

| | |
|---|---|
| **Research Impact/Earth Benefits:** | Exploration mission scenarios are increasingly engaging plant production as adjunct or major parts of life support. This is especially true for deep space and extraterrestrial surface scenarios, and there have been several lower fidelity projects that include plant production in analog tests of such scenarios. Plant growth has been a part of some simulations, and small-scale production type plant growth units are currently on the ISS. However, in none of these cases has an exploration-realistic plant growth facility been incorporated into a challenging analog site such as Antarctica to move the technology significantly toward anticipated mission structures and operations. And none incorporated sophisticated plant imaging into remote science backrooms. The EDEN ISS project is envisioned to fill that large gap, placing an ISS-inspired module near the German Antarctic Neumayer Station III. The EDEN ISS project is an active, funded project that involves a wide international team and is led by the German Aerospace Center, DLR. We have been invited to contribute our presence, expertise, and imaging technology to EDEN ISS. Opportunities for non-European participation at Neumayer III are rare, and EDEN ISS is unique in its magnitude and fidelity. This proposal seeks to seize the opportunity for important imaging technology advancement and US participation within the multinational context of the EDEN ISS project.<br>The EDEN ISS project seeks to create a structurally and operationally realistic analog to a space station, vehicular, or planetary habitat plant growth module for life support. As such, the project falls under the purview of Space Exploration and Spaceport Technology Development (SESTD) in several subcategories: In-Situ Resource Utilization Technologies (in that EDEN is dedicated to local production of food for bioregenerative Advanced Life Support), Lunar and Mars Base/Settlement Technologies (in that we will be within a robust simulation of life support technologies), and also Advanced Sensors and Detectors (in that we will be developing imaging cameras for monitoring plant health within the facility). As a working production facility, EDEN ISS will provide fresh produce to Neumayer III overwintering crews. And in the process, the project will validate key plant growth system technologies and operational protocols for use on the ISS and within future planetary bioregenerative life support systems -- a key set of technologies for long duration spaceflight and planetary habitation.<br>The project also has the potential to substantially contribute to plant health monitoring technology in terrestrial environments. Protected agriculture (production greenhouses and other crop facilities that are environmentally controlled or where climate is ameliorated) is a rapidly growing field in a world with expanding needs for food, and expanding regions of marginal crop land. Robotic tending and monitoring approaches are being embraced as time and resources saving tactics, and the type of health monitoring we are employing at EDEN ISS will contribute to the sophisticate tool box being employed in this expanding technology. |
| **Task Progress:** | There were three main developmental technology tasks for this project: NDVI photo calibration, camera /filter development, and EDEN greenhouse integration. Each of these technology tasks were applied to develop a better understanding of the biology in the greenhouse. An in-depth understanding of the NDVI process and equation is important to develop the technology for early detection of crop failure. NDVI (Normalized Difference Vegetation Index) traditionally uses light reflected in the red spectrum to compare to light reflected in the far-red spectrum. By installing a camera capable of this study, we are better able to predict crop health in the EDEN greenhouse.<br>While many different filters were tested, an NDVI-7 filter was selected for this project that was available from IRPro. This filter allows NDVI photos to be created from a single image, as opposed to the traditional method of combining a red wavelength and a near infrared wavelength photo. Photos created using the NDVI-7 filter on the GoPro camera were then processed using ImageJ and a Photo Monitoring Plug-in developed by Ned Horning for the Public Lab. This plug-in uses the single image to create an NDVI image, by utilizing the blue wavelengths to serve as the traditional red (or white) light part of the equation and the red wavelengths to serve as the near infrared part of the equation NDVI=(NIR-Red)/(NIR+Red). By using a filter that cuts most of the red light from reaching the photo sensor, it permits the red sensors to receive near infrared light, which can then be used in image creating.<br>In addition to the modified GoPro camera, the addition of the CamDo Blink system allowed us to take pictures on a selected schedule. While powered with a USB cord, the only limiting part of the system is the space on the SD card (which could hold over 25,000 photos). When CamDo activates the camera, it includes a small amount of code for scheduling and operations. The activated code causes the GoPro to search for a local wireless network and then upload the photos using an FTP (File Transfer Protocol). This allows for an autonomous data collection within the greenhouse.<br>The greenhouse integration also imposed the challenge to light spectrum differences. While traditional NDVI photos are captured with the full range of sunlight, this full spectrum is rarely available within closed habits with artificial lighting. When NDVI approaches are utilized within a greenhouse, the artificial spectrum must be adjusted to include the necessary wavelengths, and this past year (pre deployments) spent a substantial amount of time in calibrations and testing spectra.<br>After arrival in Antarctica, the PI team worked with the DLR EDEN ISS Team to prepare the FEG to activate growing systems and data collection systems. The UFSI cameras were installed in select locations to facilitate the eventual observation of a tall crop (tomato) and of a low, leafy crop (arugula or lettuce). The cameras were then calibrated for focus and image collection--first with graphic targets, and then with seedling plants. Also tested at this stage was the ability of the images to be downloaded to the server, and delivered to the UF laboratory for processing.<br>The two UF Imagers in the FEG (Future Exploration Greenhouse) capture photos twice a day. During each image capture, the photos are uploaded to the local server in Antarctica. These images are then transferred once a day to a server at DLR (German Space Agency in Bremerhaven), which is accessible over the internet. Once a day UF programs download the most recent images to UF servers for analysis. Images are received on a 24 hour delay from capture. Images are then manually processed using ImageJ software. NDVI images and graphs are uploaded for DLR access. The systems are working well; for example, returned images allowed us to detect an anomaly (which was later determined to be an error in the nutrient delivery system) before plant damage was visible to the naked eye, or the traditional monitoring cameras. |
| **Bibliography Type:** | Description: (Last Updated: 11/27/2022) |
| **Significant Media Coverage** | Wilhelm M. "Antarctic Veggies: Practice For Growing Plants On Other Planets. NPR article about the EDEN project." NPR's The Salt, April 18, 2018. https://www.npr.org/sections/thesalt/2018/04/18/601654780/antarctic-veggies-practice-for-growing-plants-on-other-planets?utm_campaign=storyshare&utm_source=twitter.com&utm_medium=soci , Apr-2018 |
| **Significant Media Coverage** | Women in Science. "Twitter post on Women in Science, including clip of Dr. Anna-Lisa Paul's research on this EDEN project in Antarctica." Twitter post, Women in Science, March 2018: https://twitter.com/ISS_Research/status/962796712241672192 , Mar-2018 |